

RESEARCH HIGHLIGHT

Automatic Alignment of Freeform USTP Systems via Physics-guided Multi-objective Optimization

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REPRESENTATIVE PUBLICATION

Jiang Li, Duo Xu, Changshuai Fang, Zexiao Li, and Xiaodong Zhang, "Sensitivity-informed multi-objective adaptive weighting method for automatic alignment of freeform USTP systems," Optics Express, published online. Duo Xu is listed as a co-first author.

PROBLEM AND MOTIVATION

Freeform ultra-short-throw projection (USTP) systems are compact but difficult to align. In this work, the freeform mirror is the main adjustable element, and its **six-degree-of-freedom pose errors** simultaneously affect geometric distortion and modulation transfer function (MTF). This coupling makes manual adjustment slow, operator-dependent, and difficult to reproduce.

The key observation is that the dominant alignment objective changes with the error scale. Large initial errors are mainly reflected by geometric distortion, while fine alignment depends more on image sharpness and MTF. A fixed metric weight is therefore not well suited to the whole alignment process.

METHOD AND CONTRIBUTIONS

I participated in building an **image-quality-driven optimization problem** workflow that connects optical simulation, structured target imaging, distortion/MTF extraction, adaptive weighting, and closed-loop 6-DOF motion execution.

- Built a Zemax non-sequential model matched to the physical prototype and analyzed full-parameter 6-DOF pose-error sensitivity.
- Designed a structured target and image-processing workflow to extract distortion modes and 93 lp/mm MTF from captured images.
- Implemented a **physics-guided adaptive weighting optimizer** with stage-wise metric scheduling and local DOF-distortion grouping.
- Validated the method on a real USTP alignment platform using industrial-camera feedback and 6-DOF motion control.

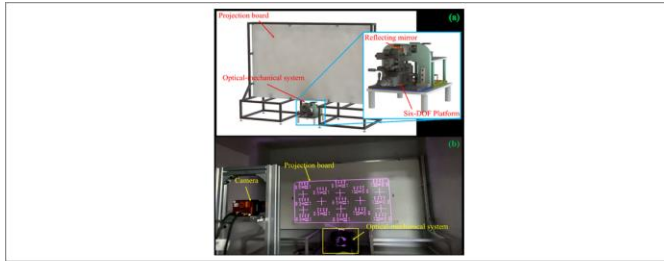
KEY RESULTS

- **Final mean MTF exceeded 70%** in repeated closed-loop alignment experiments.
- Each alignment run took **about 10 minutes** under different initial poses.
- Composite evaluation values were concentrated across 10 trials, indicating improved robustness and repeatability.
- Compared with manual alignment, the automatic method produced clearer high-frequency line-pair regions and more stable geometric boundaries.

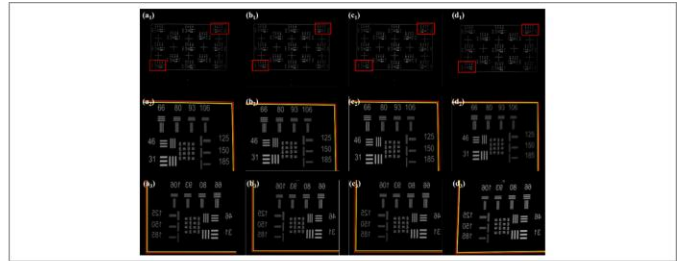
RELEVANCE TO PHD RESEARCH

This work shaped my interest in **physics-guided image analysis, computational imaging, and learning-based optimization** for real optical systems. It also gave me practical experience in connecting image feedback with optical modeling and physical actuation, which I hope to extend toward learning-based imaging and vision research during PhD study.

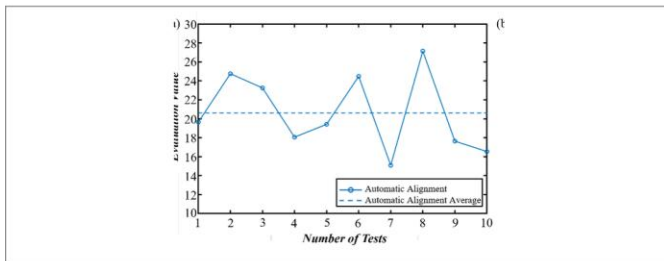
VISUAL EVIDENCE



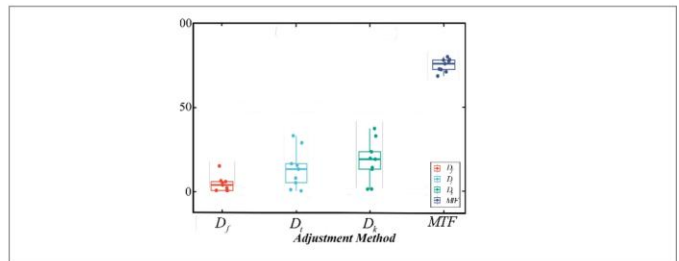
A. Experimental platform and structured target image



B. Local image-quality comparison after alignment



C. Robustness across ten automatic alignment trials



D. Statistical comparison of adjustment metrics